

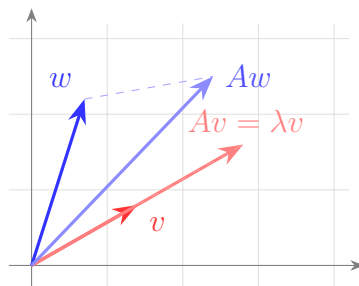
Exercise Sheet 11 – Solutions

Linear Algebra

What do the terms mean?

- **Eigenvalue λ and eigenvector v :** a number λ and a vector $v \neq 0$ with $Av = \lambda v$. That means: A only *stretches* the vector v (by the factor λ), without changing its direction.
- **Characteristic polynomial $\chi_A(t) = \det(A - tI)$:** the roots of this polynomial are exactly the eigenvalues. (Reason: $Av = \lambda v$ means $(A - \lambda I)v = 0$ with $v \neq 0$, so $A - \lambda I$ is *not* invertible, hence $\det(A - \lambda I) = 0$.)
- **Eigenspace $E_\lambda = \ker(A - \lambda I)$:** all eigenvectors for λ (plus the zero vector). You find it by solving the system $(A - \lambda I)v = 0$.
- **Algebraic multiplicity $\text{AM}(\lambda)$:** how often λ appears as a root of the char. polynomial (the exponent of $(t - \lambda)$).
- **Geometric multiplicity $\text{GM}(\lambda)$:** $\dim E_\lambda =$ number of basis vectors of the eigenspace = number of free variables when solving $(A - \lambda I)v = 0$.
- **It always holds that $1 \leq \text{GM}(\lambda) \leq \text{AM}(\lambda)$.** If all $\text{GM} = \text{AM}$, then A is *diagonalizable*.

[Graphical explanation – not part of the solution]



*The basic idea: multiplying a vector by A usually rotates and stretches it (blue: $w \rightarrow Aw$ points elsewhere). An **eigenvector** (red) is a special direction that A only stretches – Av lies on the same line as v . The stretch factor is the eigenvalue λ .*

Exercise 43

Problem. For $A = \begin{pmatrix} 4 & 1 & 1 \\ 2 & 5 & 2 \\ -1 & -1 & 2 \end{pmatrix}$ and $B = \begin{pmatrix} 5 & 2 & 3 \\ 2 & 4 & 2 \\ -3 & -2 & -1 \end{pmatrix}$ determine the characteristic polynomial, the eigenvalues with algebraic and geometric multiplicity, and a basis of each eigenspace – in **two ways**.

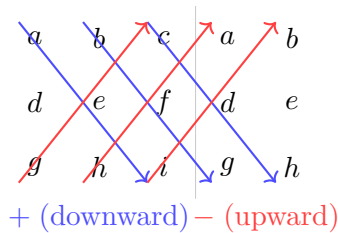
What needs to be done?

- **Method 1:** multiply out $\chi_A(t) = \det(A - tI)$ with the **Sarrus rule**, then factor with the **theorem on integer roots** (candidates = divisors of the constant term).
- **Method 2:** use **row/column operations** to factor out a term $(\lambda - t)$ early – this saves the multiplying-out.
- Then, per eigenvalue λ , solve the system $(A - \lambda I)v = 0 \Rightarrow$ basis of the eigenspace and GM.

Matrix A

Method 1: Sarrus + integer roots.

[Graphical explanation – not part of the solution]



Sarrus rule: write the first two columns again on the right. The three diagonals going down (blue) count +, the three going up (red) count -. Formula: $\det = aei + bfg + cdh - ceg - afh - bdi$.

Step 1: Write down $A - tI$. tI is the diagonal matrix with t all along the diagonal. So subtracting it changes *only the diagonal* of A :

$$A - tI = \begin{pmatrix} 4 & 1 & 1 \\ 2 & 5 & 2 \\ -1 & -1 & 2 \end{pmatrix} - \begin{pmatrix} t & 0 & 0 \\ 0 & t & 0 \\ 0 & 0 & t \end{pmatrix} = \begin{pmatrix} 4-t & 1 & 1 \\ 2 & 5-t & 2 \\ -1 & -1 & 2-t \end{pmatrix}.$$

Step 2: Apply Sarrus.

The rule in general. For an arbitrary 3×3 matrix with letters:

$$\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} \longrightarrow \det = aei + bfg + cdh - ceg - afh - bdi.$$

(The first three products = diagonals going down (+), the last three = diagonals going up (-), see the figure.)

Read off the letters from our matrix. From $A - tI = \begin{pmatrix} 4-t & 1 & 1 \\ 2 & 5-t & 2 \\ -1 & -1 & 2-t \end{pmatrix}$ we read entry by entry:

$$\begin{aligned} a &= 4 - t, & b &= 1, & c &= 1, \\ d &= 2, & e &= 5 - t, & f &= 2, \\ g &= -1, & h &= -1, & i &= 2 - t. \end{aligned}$$

Compute the six products one by one:

$$\begin{aligned} aei &= (4 - t)(5 - t)(2 - t), & ceg &= 1 \cdot (5 - t) \cdot (-1) = -(5 - t), \\ bfg &= 1 \cdot 2 \cdot (-1) = -2, & afh &= (4 - t) \cdot 2 \cdot (-1) = -2(4 - t), \\ cdh &= 1 \cdot 2 \cdot (-1) = -2, & bdi &= 1 \cdot 2 \cdot (2 - t) = 2(2 - t). \end{aligned}$$

Substitute into the formula (first three with +, last three with -):

$$\begin{aligned}\det(A - tI) &= aei + bfg + cdh - ceg - afh - bdi \\ &= (4 - t)(5 - t)(2 - t) + (-2) + (-2) - (- (5 - t)) - (- 2(4 - t)) - 2(2 - t) \\ &= (4 - t)(5 - t)(2 - t) - 2 - 2 + (5 - t) + 2(4 - t) - 2(2 - t).\end{aligned}$$

The bracket term multiplied out: $(4 - t)(5 - t)(2 - t) = -t^3 + 11t^2 - 38t + 40$. The remaining terms: $-2 - 2 + 5 - t + 8 - 2t - 4 + 2t = 5 - t$. Together:

$$\chi_A(t) = -t^3 + 11t^2 - 38t + 40 + 5 - t = -t^3 + 11t^2 - 39t + 45.$$

[Explanation added] Quick check: constant term $\chi_A(0) = 45 = \det A \checkmark$, and the t^2 -coefficient $11 = \text{trace}(A) = 4 + 5 + 2 \checkmark$.

Step 3: Factor. Factoring out -1 : $\chi_A(t) = -(t^3 - 11t^2 + 39t - 45)$. Integer roots are divisors of 45 – trying them shows: $t = 3$ and $t = 5$ give 0. The third root comes from the trace ($3 + 5 + \lambda_3 = 11$): $\lambda_3 = 3$. So the roots are **3, 3, 5**.

From roots to factored form. Each root λ becomes a factor $(t - \lambda)$ – a double root accordingly becomes a square:

$$t^3 - 11t^2 + 39t - 45 = (t - 3) \underbrace{(t - 3)}_{\text{the 3 twice}} (t - 5) = (t - 3)^2(t - 5).$$

We pull the -1 from before into the last factor, and since $(t - 3)^2 = (3 - t)^2$ and $-(t - 5) = (5 - t)$, this becomes:

$$\boxed{\chi_A(t) = -(t - 3)^2(t - 5) = (3 - t)^2(5 - t).}$$

Exponent = algebraic multiplicity. The AM is simply how often the root occurs = the exponent of the factor: $\lambda = 3$ is squared $\Rightarrow$ AM 2; $\lambda = 5$ appears once $\Rightarrow$ AM 1.

Method 2: Factor out early (column trick).

[Graphical explanation – not part of the solution]

$$\begin{pmatrix} 4-t & 1 & 1 \\ 2 & 5-t & 2 \\ -1 & -1 & 2-t \end{pmatrix} \xrightarrow{R_1 + R_2 + R_3} \begin{pmatrix} 5-t & 5-t & 5-t \\ 2 & 5-t & 2 \\ -1 & -1 & 2-t \end{pmatrix}$$

Observation: in A every column sums to 5 (e.g. $4 + 2 - 1 = 5$). So in $A - tI$ every column sums to $5 - t$. If you add all rows together ($R_1 \rightarrow R_1 + R_2 + R_3$), the top row becomes all $(5 - t)$ – which you can pull out as a factor.

Step 1: $R_1 \rightarrow R_1 + R_2 + R_3$ (does not change the determinant). We add the three rows of $A - tI$ and write the result into the first row – column by column:

$$\text{Column 1: } (4 - t) + 2 + (-1) = 5 - t,$$

$$\text{Column 2: } 1 + (5 - t) + (-1) = 5 - t,$$

$$\text{Column 3: } 1 + 2 + (2 - t) = 5 - t.$$

So the new first row is $(5 - t, 5 - t, 5 - t)$, the other two stay. Now factor out $(5 - t)$:

$$\det(A - tI) = \det \begin{pmatrix} 5-t & 5-t & 5-t \\ 2 & 5-t & 2 \\ -1 & -1 & 2-t \end{pmatrix} = (5-t) \det \begin{pmatrix} 1 & 1 & 1 \\ 2 & 5-t & 2 \\ -1 & -1 & 2-t \end{pmatrix}.$$

Step 2: columns $C_2 \rightarrow C_2 - C_1$ **and** $C_3 \rightarrow C_3 - C_1$ (subtract column 1 from columns 2 and 3 to get zeros in the upper right). Entry by entry:

$$C_2 - C_1 : 1 - 1 = 0, \quad (5 - t) - 2 = 3 - t, \quad -1 - (-1) = 0 \Rightarrow \text{column } (0, 3 - t, 0),$$

$$C_3 - C_1 : 1 - 1 = 0, \quad 2 - 2 = 0, \quad (2 - t) - (-1) = 3 - t \Rightarrow \text{column } (0, 0, 3 - t).$$

$$= (5 - t) \det \begin{pmatrix} 1 & 0 & 0 \\ 2 & 3 - t & 0 \\ -1 & 0 & 3 - t \end{pmatrix} = (5 - t)(3 - t)(3 - t) = (5 - t)(3 - t)^2.$$

[Explanation added] The last matrix is (lower) triangular $\Rightarrow$ determinant = product of the diagonal $1 \cdot (3 - t) \cdot (3 - t)$.

Same result as Method 1: $\chi_A(t) = (5 - t)(3 - t)^2$, eigenvalues $\lambda = 5$ (AM 1), $\lambda = 3$ (AM 2).

Eigenspaces.

$\lambda = 5$: solve $(A - 5I)v = 0$. First $A - 5I$ (only the diagonal decreases by 5):

$$A - 5I = \begin{pmatrix} 4 & 1 & 1 \\ 2 & 5 & 2 \\ -1 & -1 & 2 \end{pmatrix} - \begin{pmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{pmatrix} = \begin{pmatrix} -1 & 1 & 1 \\ 2 & 0 & 2 \\ -1 & -1 & -3 \end{pmatrix}.$$

Then Gauss:

$$\begin{pmatrix} -1 & 1 & 1 \\ 2 & 0 & 2 \\ -1 & -1 & -3 \end{pmatrix} \xrightarrow[R_3 \rightarrow R_3 - R_1]{R_2 \rightarrow R_2 + 2R_1} \begin{pmatrix} -1 & 1 & 1 \\ 0 & 2 & 4 \\ 0 & -2 & -4 \end{pmatrix} \xrightarrow{R_3 \rightarrow R_3 + R_2} \begin{pmatrix} -1 & 1 & 1 \\ 0 & 2 & 4 \\ 0 & 0 & 0 \end{pmatrix}.$$

Each row of the echelon form is an equation for (x, y, z) – entries times x, y, z , equal to 0: $(0, 2, 4) \Rightarrow 2y + 4z = 0$, $(-1, 1, 1) \Rightarrow -x + y + z = 0$. The free variable is z ; solving: $2y + 4z = 0 \Rightarrow y = -2z$, then $-x + y + z = 0 \Rightarrow x = y + z = -z$. With $z = 1$:

$$E_5 = \text{span}\{(-1, -2, 1)\}, \quad \text{GM}(5) = 1.$$

$\lambda = 3$: First $A - 3I$:

$$A - 3I = \begin{pmatrix} 4 & 1 & 1 \\ 2 & 5 & 2 \\ -1 & -1 & 2 \end{pmatrix} - \begin{pmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ -1 & -1 & -1 \end{pmatrix}.$$

Then Gauss:

$$\begin{pmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ -1 & -1 & -1 \end{pmatrix} \xrightarrow[R_3 \rightarrow R_3 + R_1]{R_2 \rightarrow R_2 - 2R_1} \begin{pmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

The only nontrivial row $(1, 1, 1)$ means $x + y + z = 0$ (entries times x, y, z). Free variables y, z , so $x = -y - z$. With $(y, z) = (1, 0)$ and $(0, 1)$:

$$E_3 = \text{span}\{(-1, 1, 0), (-1, 0, 1)\}, \quad \text{GM}(3) = 2.$$

$$\chi_A(t) = (5 - t)(3 - t)^2.$$

$\lambda = 5$: AM 1, GM 1, basis $\{(-1, -2, 1)\}$. $\lambda = 3$: AM 2, GM 2, basis $\{(-1, 1, 0), (-1, 0, 1)\}$.

All GM = AM $\Rightarrow A$ is diagonalizable.

Matrix B

Method 1: Sarrus. First $B - tI$ (again only the diagonal):

$$B - tI = \begin{pmatrix} 5 & 2 & 3 \\ 2 & 4 & 2 \\ -3 & -2 & -1 \end{pmatrix} - \begin{pmatrix} t & 0 & 0 \\ 0 & t & 0 \\ 0 & 0 & t \end{pmatrix} = \begin{pmatrix} 5-t & 2 & 3 \\ 2 & 4-t & 2 \\ -3 & -2 & -1-t \end{pmatrix}.$$

Read off the letters (same formula $\det = aei + bfg + cdh - ceg - afh - bdi$ as for A):

$$\begin{aligned} a &= 5-t, & b &= 2, & c &= 3, \\ d &= 2, & e &= 4-t, & f &= 2, \\ g &= -3, & h &= -2, & i &= -1-t. \end{aligned}$$

The six products:

$$\begin{aligned} aei &= (5-t)(4-t)(-1-t), & ceg &= 3 \cdot (4-t) \cdot (-3) = -9(4-t), \\ bfg &= 2 \cdot 2 \cdot (-3) = -12, & afh &= (5-t) \cdot 2 \cdot (-2) = -4(5-t), \\ cdh &= 3 \cdot 2 \cdot (-2) = -12, & bdi &= 2 \cdot 2 \cdot (-1-t) = 4(-1-t). \end{aligned}$$

Substitute into the formula:

$$\begin{aligned} \chi_B(t) &= aei + bfg + cdh - ceg - afh - bdi \\ &= (5-t)(4-t)(-1-t) + (-12) + (-12) - (-9(4-t)) - (-4(5-t)) - 4(-1-t) \\ &= (5-t)(4-t)(-1-t) - 12 - 12 + 9(4-t) + 4(5-t) - 4(-1-t) \\ &= (5-t)(4-t)(-1-t) + 36 - 9t = -t^3 + 8t^2 - 20t + 16. \end{aligned}$$

[Explanation added] Check: $\chi_B(0) = 16 = \det B \checkmark$, t^2 -coefficient $8 = \text{trace}(B) = 5 + 4 - 1 \checkmark$.

Factor. $\chi_B(t) = -(t^3 - 8t^2 + 20t - 16)$. Integer roots can only be divisors of 16 ($\pm 1, \pm 2, \pm 4, \dots$). You try them out – *testing* means: plug in for t and see whether 0 comes out:

$$t = 2 : 8 - 32 + 40 - 16 = 0 \checkmark, \quad t = 4 : 64 - 128 + 80 - 16 = 0 \checkmark.$$

2 and 4 give 0, so they are roots. The **third** one (still unknown, call it λ_3) comes for free: the three roots sum to 8 (the trace), and $2 + 4 + \lambda_3 = 8$ gives $\lambda_3 = 2$. So the roots are **2, 2, 4**. Each root becomes a factor, the double 2 a square:

$$\boxed{\chi_B(t) = -(t-2)^2(t-4) = (2-t)^2(4-t)}.$$

Exponent = AM: $\lambda = 2$ squared $\Rightarrow$ AM 2; $\lambda = 4$ once $\Rightarrow$ AM 1.

Method 2: Factor out early (column trick). In B too every column sums to 4 ($5 + 2 - 3 = 4$, $2 + 4 - 2 = 4$, $3 + 2 - 1 = 4$), so in $B - tI$ to $4 - t$.

Step 1: $R_1 \rightarrow R_1 + R_2 + R_3$ (does not change the determinant). The three rows of $B - tI$ added up, column by column:

$$\begin{aligned} \text{Column 1: } & (5-t) + 2 + (-3) = 4-t, \\ \text{Column 2: } & 2 + (4-t) + (-2) = 4-t, \\ \text{Column 3: } & 3 + 2 + (-1-t) = 4-t. \end{aligned}$$

New first row $(4 - t, 4 - t, 4 - t)$, the others stay. Now factor out $(4 - t)$:

$$\det(B - tI) = \det \begin{pmatrix} 4-t & 4-t & 4-t \\ 2 & 4-t & 2 \\ -3 & -2 & -1-t \end{pmatrix} = (4-t) \det \begin{pmatrix} 1 & 1 & 1 \\ 2 & 4-t & 2 \\ -3 & -2 & -1-t \end{pmatrix}.$$

Step 2: columns $C_2 \rightarrow C_2 - C_1$ **and** $C_3 \rightarrow C_3 - C_1$ (subtract column 1 from columns 2 and 3). Entry by entry:

$$C_2 - C_1 : 1 - 1 = 0, \quad (4 - t) - 2 = 2 - t, \quad -2 - (-3) = 1 \Rightarrow \text{column } (0, 2 - t, 1),$$

$$C_3 - C_1 : 1 - 1 = 0, \quad 2 - 2 = 0, \quad (-1 - t) - (-3) = 2 - t \Rightarrow \text{column } (0, 0, 2 - t).$$

$$= (4 - t) \det \begin{pmatrix} 1 & 0 & 0 \\ 2 & 2 - t & 0 \\ -3 & 1 & 2 - t \end{pmatrix} = (4 - t)(2 - t)(2 - t) = (4 - t)(2 - t)^2.$$

[Explanation added] The last matrix is (lower) triangular $\Rightarrow$ determinant = product of the diagonal $1 \cdot (2 - t) \cdot (2 - t)$.

Same result as Method 1: $\chi_B(t) = (4 - t)(2 - t)^2$, eigenvalues $\lambda = 2$ (AM 2), $\lambda = 4$ (AM 1).

Eigenspaces.

$\lambda = 4$: First $B - 4I$:

$$B - 4I = \begin{pmatrix} 5 & 2 & 3 \\ 2 & 4 & 2 \\ -3 & -2 & -1 \end{pmatrix} - \begin{pmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 0 & 2 \\ -3 & -2 & -5 \end{pmatrix}.$$

Then Gauss:

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 0 & 2 \\ -3 & -2 & -5 \end{pmatrix} \xrightarrow[R_3 \rightarrow R_3 + 3R_1]{R_2 \rightarrow R_2 - 2R_1} \begin{pmatrix} \boxed{1} & 2 & 3 \\ 0 & \boxed{-4} & -4 \\ 0 & 4 & 4 \end{pmatrix} \xrightarrow{R_3 \rightarrow R_3 + R_2} \begin{pmatrix} \boxed{1} & 2 & 3 \\ 0 & \boxed{-4} & -4 \\ 0 & 0 & 0 \end{pmatrix}.$$

Rows as equations: $(0, -4, -4) \Rightarrow -4y - 4z = 0 \Rightarrow y = -z$; $(1, 2, 3) \Rightarrow x + 2y + 3z = 0 \Rightarrow x = -2y - 3z = -z$. With $z = 1$:

$$E_4 = \text{span}\{(-1, -1, 1)\}, \quad \text{GM}(4) = 1.$$

$\lambda = 2$: First $B - 2I$:

$$B - 2I = \begin{pmatrix} 5 & 2 & 3 \\ 2 & 4 & 2 \\ -3 & -2 & -1 \end{pmatrix} - \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix} = \begin{pmatrix} 3 & 2 & 3 \\ 2 & 2 & 2 \\ -3 & -2 & -3 \end{pmatrix}.$$

Then Gauss:

$$\begin{pmatrix} 3 & 2 & 3 \\ 2 & 2 & 2 \\ -3 & -2 & -3 \end{pmatrix} \xrightarrow{R_2 \rightarrow \frac{1}{2}R_2} \begin{pmatrix} 3 & 2 & 3 \\ 1 & 1 & 1 \\ -3 & -2 & -3 \end{pmatrix} \xrightarrow[R_3 \rightarrow R_3 + 3R_2]{R_1 \rightarrow R_1 - 3R_2} \begin{pmatrix} 0 & -1 & 0 \\ \boxed{1} & 1 & 1 \\ 0 & 1 & 0 \end{pmatrix}.$$

Rows as equations: $(0, -1, 0) \Rightarrow -y = 0$, so $y = 0$; $(1, 1, 1) \Rightarrow x + y + z = 0 \Rightarrow x = -z$. Only free variable z . With $z = 1$:

$$E_2 = \text{span}\{(-1, 0, 1)\}, \quad \text{GM}(2) = 1.$$

$$\chi_B(t) = (2 - t)^2(4 - t).$$

$\lambda = 2$: AM 2, GM 1, basis $\{(-1, 0, 1)\}$. $\lambda = 4$: AM 1, GM 1, basis $\{(-1, -1, 1)\}$.

Here $\text{GM}(2) = 1 < \text{AM}(2) = 2 \Rightarrow B$ is *not* diagonalizable.

Exercise 44

Problem. Determine the eigenvalues (with AM, GM) and bases of the eigenspaces of

$$A = \begin{pmatrix} 1 & 0 & 5 \\ 0 & 6 & 0 \\ 1 & 0 & 5 \end{pmatrix} \text{ and } B = \begin{pmatrix} 5 & 5 & 5 & 5 \\ 7 & 7 & 7 & 7 \\ 11 & 11 & 11 & 11 \\ -23 & -23 & -23 & -23 \end{pmatrix}.$$

What needs to be done?

- **No characteristic polynomial!** Instead use properties: $\lambda = 0$ exactly when A is not invertible; sum of eigenvalues = trace; product = det; $\text{GM} \leq \text{AM}$; read off obvious eigenvectors.

Matrix A

[Graphical explanation – not part of the solution]

$$A = \begin{pmatrix} 1 & 0 & 5 \\ 0 & 6 & 0 \\ 1 & 0 & 5 \end{pmatrix} \begin{array}{l} \text{row 1} \\ Ae_2 = (0, 6, 0) = 6e_2 \Rightarrow \lambda = 6 \\ \text{row 3} \Rightarrow \det = 0 \Rightarrow \lambda = 0 \end{array}$$

Two things are immediate: (1) row 1 = row 3, so A is singular $\Rightarrow 0$ is an eigenvalue. (2) The middle column is $(0, 6, 0)$, i.e. A sends $e_2 = (0, 1, 0)$ to $6e_2 \Rightarrow 6$ is an eigenvalue with eigenvector e_2 .

Step 1: $\lambda = 0$. Row 1 = row 3, and two equal rows give $\det A = 0$. So A is not invertible, i.e. there is $v \neq 0$ with $Av = 0 = 0 \cdot v - \lambda v = 0$ is an eigenvalue, and the eigenspace is $\ker A$. The rows of $Av = 0$ as equations:

$$\text{Row 2 } (0, 6, 0) : 6y = 0 \Rightarrow y = 0; \quad \text{Row 1 } (1, 0, 5) : x + 5z = 0 \Rightarrow x = -5z.$$

z is free; with $z = 1$:

$$E_0 = \text{span}\{(-5, 0, 1)\}, \quad \text{GM}(0) = 1.$$

Step 2: $\lambda = 6$. The middle column is $(0, 6, 0)$, i.e. $Ae_2 = 6e_2$ – so 6 is an eigenvalue with eigenvector e_2 . The whole eigenspace is $\ker(A - 6I)$:

$$A - 6I = \begin{pmatrix} 1 & 0 & 5 \\ 0 & 6 & 0 \\ 1 & 0 & 5 \end{pmatrix} - \begin{pmatrix} 6 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 6 \end{pmatrix} = \begin{pmatrix} -5 & 0 & 5 \\ 0 & 0 & 0 \\ 1 & 0 & -1 \end{pmatrix}.$$

Rows as equations: $(-5, 0, 5) \Rightarrow -5x + 5z = 0 \Rightarrow x = z$; $(1, 0, -1) \Rightarrow x - z = 0$ (the same). No condition on y , so y and z are free. With $(y, z) = (1, 0)$ and $(0, 1)$:

$$E_6 = \text{span}\{(0, 1, 0), (1, 0, 1)\}, \quad \text{GM}(6) = 2.$$

Step 3: Multiplicities. A 3×3 matrix has 3 eigenvalues in total (with multiplicity), so the AMs sum to 3. Since $\text{GM} \leq \text{AM}$, we have $\text{AM}(6) \geq \text{GM}(6) = 2$, and 0 is an eigenvalue, so $\text{AM}(0) \geq 1$. Already $2 + 1 = 3$ – everything is used up: $\text{AM}(6) = 2$, $\text{AM}(0) = 1$. *[Explanation added]* Check with the trace: $0 \cdot 1 + 6 \cdot 2 = 12 = \text{trace}(A) = 1 + 6 + 5$ ✓.

$\lambda = 0$: AM 1, GM 1, basis $\{(-5, 0, 1)\}$. $\lambda = 6$: AM 2, GM 2, basis $\{(0, 1, 0), (1, 0, 1)\}$. All GM = AM $\Rightarrow A$ is diagonalizable.

Matrix B

[Graphical explanation – not part of the solution]

$$B = \begin{pmatrix} 5 & 5 & 5 & 5 \\ 7 & 7 & 7 & 7 \\ 11 & 11 & 11 & 11 \\ -23 & -23 & -23 & -23 \end{pmatrix} \left\{ \begin{array}{l} \text{every row} = c_i \cdot (1, 1, 1, 1) \\ c = (5, 7, 11, -23) \\ \Rightarrow \text{rank } 1 \end{array} \right.$$

B has identical columns everywhere – every row is a multiple of $(1, 1, 1, 1)$. Such a matrix has **rank** 1: all rows lie “on one line”. Hence the kernel is huge (dimension $4 - 1 = 3$) and 0 is an eigenvalue.

Step 1: $\lambda = 0$. Every row of B is a multiple of $(1, 1, 1, 1)$ – so there is only *one* independent row, $\text{rank}(B) = 1$. By the dimension formula $\dim \ker(B) = 4 - \text{rank} = 4 - 1 = 3$. A kernel larger than $\{0\}$ means: there is $v \neq 0$ with $Bv = 0$, so 0 is an eigenvalue with $\text{GM}(0) = \dim \ker = 3$.

The kernel: each row of $Bv = 0$ is (e.g. row 1) $5a + 5b + 5c + 5d = 5(a + b + c + d) = 0$, i.e. $a + b + c + d = 0$. One equation, four unknowns $\Rightarrow$ three free variables (b, c, d ; $a = -b - c - d$):

$$E_0 = \text{span}\{(-1, 1, 0, 0), (-1, 0, 1, 0), (-1, 0, 0, 1)\}, \quad \text{GM}(0) = 3.$$

Step 2: $\text{AM}(0) = 4$. B is 4×4 , so it has 4 eigenvalues (with multiplicity). Three of them are 0 (since $\text{AM}(0) \geq \text{GM}(0) = 3$). Call the fourth μ . The sum of all eigenvalues is the trace: $5 + 7 + 11 - 23 = 0$, so $0 + 0 + 0 + \mu = 0 \Rightarrow \mu = 0$. So 0 is the only eigenvalue:

$$\text{AM}(0) = 4.$$

[Explanation added] Put differently: a rank-1 matrix $cr^\top$ has only the eigenvalue $r^\top c = \text{trace}$ (once) and otherwise 0. Here $\text{trace} = 0$, so *all* eigenvalues are 0 (the matrix is nilpotent, $B^2 = 0$).

$\lambda = 0$: AM 4, GM 3, basis $\{(-1, 1, 0, 0), (-1, 0, 1, 0), (-1, 0, 0, 1)\}$. $\text{GM}(0) = 3 < \text{AM}(0) = 4 \Rightarrow B$ is <i>not</i> diagonalizable.
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Exercise 45

Problem. $A, B \in \mathbb{F}^{n \times n}$. Prove the following claims.

What needs to be done?

- In each claim we start with “ λ is an eigenvalue of A ” = “there is $v \neq 0$ with $Av = \lambda v$ ” and check that the same v is also an eigenvector of the new matrix.

(a) If λ is an eigenvalue of A and $c \in \mathbb{F}$, then $\lambda + c$ is an eigenvalue of $A + cI$.

Proof. Let $v \neq 0$ with $Av = \lambda v$. Then

$$(A + cI)v = Av + cv = \lambda v + cv = (\lambda + c)v.$$

Since $v \neq 0$, $\lambda + c$ is an eigenvalue of $A + cI$ (with the same eigenvector v). $\square$

(b) If λ is an eigenvalue of A , then λ^2 is an eigenvalue of A^2 .

Proof. Let $v \neq 0$ with $Av = \lambda v$. Then

$$A^2v = A(Av) = A(\lambda v) = \lambda(Av) = \lambda(\lambda v) = \lambda^2v. \quad \square$$

(c) If A is invertible and λ is an eigenvalue of A , then λ^{-1} is an eigenvalue of A^{-1} .

Proof. Since A is invertible, 0 is not an eigenvalue (otherwise there would be $v \neq 0$ with $Av = 0$, so A would not be injective). Hence $\lambda \neq 0$. From $Av = \lambda v$, applying A^{-1} :

$$v = A^{-1}(\lambda v) = \lambda A^{-1}v \implies A^{-1}v = \frac{1}{\lambda}v.$$

Since $v \neq 0$, λ^{-1} is an eigenvalue of A^{-1} . $\square$

(d) If v is an eigenvector of both A and B , then v is also an eigenvector of AB .

Proof. Let $v \neq 0$ with $Av = \lambda v$ and $Bv = \mu v$. Then

$$(AB)v = A(Bv) = A(\mu v) = \mu(Av) = \mu(\lambda v) = (\lambda\mu)v.$$

Since $v \neq 0$, v is an eigenvector of AB for the eigenvalue $\lambda\mu$. $\square$

Overview: Results 43 & 44

Matrix	Eigenvalue (λ)	AM, GM	Basis of the eigenspace
43 A	3	2, 2	$(-1, 1, 0), (-1, 0, 1)$
	5	1, 1	$(-1, -2, 1)$
43 B	2	2, 1	$(-1, 0, 1)$
	4	1, 1	$(-1, -1, 1)$
44 A	0	1, 1	$(-5, 0, 1)$
	6	2, 2	$(0, 1, 0), (1, 0, 1)$
44 B	0	4, 3	$(-1, 1, 0, 0), (-1, 0, 1, 0), (-1, 0, 0, 1)$

Diagonalizable are 43 A and 44 A (all GM = AM). *Not diagonalizable* are 43 B and 44 B (one GM < AM each).